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# Gaia: ESA's map of the Galaxy

Gaia was a European Space Agency space observatory, launched on 19 December 2013. From a post far beyond the Moon, it measured the positions, motions, brightness, and colors of nearly two billion stars and other objects across the Milky Way and beyond.

That is a huge census of the brighter stars, not a complete map of every star. The Milky Way holds a few hundred billion stars; many are too faint, or hidden by dust, for Gaia to see.

The mission was planned for five years of science. Those five years ran from July 2014 through the end of June 2019. Observations then continued until 15 January 2025, about 10.5 years in all, until the cold gas that kept the spacecraft spinning ran out. After a short engineering period, Gaia was switched off on 27 March 2025 and left in an orbit around the Sun.

The data are still being reduced. Data Release 4, from the first 5.5 years, is due on 2 December 2026. The final catalogue, Data Release 5, will use the whole 10.5 years and is not expected before the end of 2030.

The point of the measurements was distance and motion. As Earth goes around the Sun, a nearby star appears to shift slightly against more distant stars. That tiny shift is the star's parallax. How small those angles are is on The angle ladder.

## What Gaia found that no one expected

In 2018, Gaia showed that a large part of the Milky Way's inner halo came from another galaxy, now called Gaia-Enceladus, that merged with ours about 10 billion years ago. Its stars still move together the wrong way, like debris from an old collision.

That same year, Gaia found a snail-shell spiral in the motions of nearby stars. The Galaxy's disk is still wobbling, like a pond after a stone hits it, probably because the Sagittarius dwarf galaxy passed through it.

In 2024, Gaia caught a star swaying around something invisible. That was Gaia BH3, a sleeping black hole about 33 times the Sun's mass and about 1,926 light-years away. It is the most massive star-sized black hole found in our galaxy, and the third such black hole Gaia found. The first two, Gaia BH1 and Gaia BH2, came out of Data Release 3 and were confirmed in 2023.

## Key facts

- Gaia was built for the European Space Agency and launched on 19 December 2013.
- Science was planned for five years, from July 2014 through June 2019, and ran about 10.5 years, until 15 January 2025.
- It measured nearly two billion stars and other objects, not every star in the Milky Way.
- Data Release 4 is due on 2 December 2026. The final catalogue is not expected before the end of 2030.
- Unexpected findings include the Gaia-Enceladus merger, a snail-shell wobble in the disk, and the sleeping black holes BH1, BH2, and BH3.

Sources: ESA, Gaia; the mission extension, end of observations, and data-release plan; ESA on BH1 and BH2; and ESA on BH3, announced 16 April 2024.